How to Assist Online Instructors in K-12 Education in China 如何辅助中国K-12学校教师实施在线教育

2022-08-31

(Peer Review)Yupei Duan-Final Research Proposal

( ) - K-12

How to Assist Online Instructors in K-12 Education in China

Yupei Duan

Curriculum Center, Future Leadership Academy, Hai Nan, China

School of Information and Learning Technology, University of Missouri, Missouri, The USA

December 12, 2020

How to Assist Online Instructors in K-12 Education in China

Background and Significance of the Research

Carliner and Shank (2008) have summarized four key challenges of e-learning: Organizational Barriers, Pedagogical, Technical and Financial Issues. China is faced with most of these issues, together with some unique challenges (Y. Wang et al., 2018a). Teachers are vital component in education. Online teachers have a variety of roles they need to play, such as: evaluator, administrator, technologist, advisor/counselor, researcher, etc. and they should be equipped with multiple competencies (Baran et al., 2011). It is critical to prepare and support teachers for online teaching so that they know what to expect and how to establish their online teacher persona through online pedagogies, and also develop positive attitudes towards online teaching. By incorporating collaborative work groups, community building, and group discussions into professional development programs, and sustaining their continuity, teachers will have an opportunity to participate in communities of practice and transform their teaching by socially constructing their knowledge and practices (King, 2002).

The COVID-19 affected the whole world a lot. It was a test for the online education in different countries and districts. During the pandemic period, educators and students had to rely on online education to continue schoolings. The development of Online education in China has been constrained by a lot of limitations. Zhanyuan Du, Vice Minister of Education of China, has stressed that there are three key tasks for e-learning development in China: “teachers’ and students’ acknowledgement of e-learning, teachers’ capacity for integrating ICT with daily instructions, and production of sufficient quality resources so that teachers can focus on pedagogical design (– – , 2015).” While more teachers in China are equipped with the basic Information and Communication Technology (ICT) knowledge and skills and get more knowledge about e-learning, the situation of China’s online education might be better. Since ICT is the foundation of online education, teachers who have more abilities about ICT, who could understand the value and usability of online education, who might integrate ICT more in their daily instructions, and who have more possibilities to yield higher-quality online education production.

Problem Statement

This study aims to explore the experiences, suggestions and standards for assisting online instructors to present higher-quality online instructions drawing mostly from the USA, and introduces some useful strategies from institutes who have been endeavoring to establish high standard online education from all over the world to China’s K-12 schools. With practice and comparation to make the most suitable one for China.

With collecting data from practices in some volunteer Chinese K-12 schools, the researcher wants to finish the first draft of Chinese K-12 online education standards and design some useful in-service or pre-service trainings for Chinese online instructors.

Purpose of the Study

The researcher will focus on the online education at China’ K-12 school context among the different types of online education. The COVID-19 pandemic has made all stake holders at schools: parents, students, educators and administrators to realize the value of high quality of online education much better. There are a lot of startups and E-giant companies are seizing the profit of online education by inventing new Apps and platforms, and establishing online courses and commercial training programs to the K-12 students in China. There has been so much growth in this market that the government has issued a lot of regulations to control them(*Growth of China’s Online Education Industry Spurs New Regulations*, 2020) . The fast increase need of high-quality online education from learners are apparent. Yet the supportive standards for evaluating and establishing online education were limited in China. On the other hand. If the students can get high quality online education from their own school with their familiar educators and yield more learning outcomes, why they need to pay extra money out of school? The more expansive thing for the learners is not money but time.

The author could not find formal instructions to assist teachers about how to do high-quality online teaching in Chinese. However, there is a great deal of information in English. What kinds of online teaching standards, teaching strategies suggestions the Chinese online educators need currently? What kinds of online learning tips Chinese learners should be aware of to yield higher online learning outcomes? Is the information workable, effective and efficient? The mixed-methods will be used during the research.

Literature Review

Sources deemed most relevant to the topics were selected for further analysis; preference was made to recent (published during 1999-2020) articles, but older sources were included especially the classical ones which contain impressive points and suggestions. These articles were searched and collected through google scholar and University of Missouri Columbia Libraries, utilizing the following search tactics: K-12, Online Education, Online Teaching, Standards, etc.

Online Education Definition and its Development in China

E-learning, also known as online learning, digital learning or computer-based learning, can be defined as education provided on digital devices that support learning (Clark & Mayer, 2011). The use of Information and Communication Technology (ICT) in education has given rise to diversified pedagogical models and methods, including networked learning, multimedia education, online and open education, and blended learning. In China, the term 'ICT in Education' is used interchangeably with the more general term 'e-learning' (Y. Wang et al., 2018b). With rapid economic development, China is becoming able to provide better infrastructure and other necessary conditions for schools (e.g. more high-quality computer labs, more learning soft wares and more online learning strategies), E-learning is believed to be a promising approach since it offers students ways to interact with experienced teachers or professors (Q. Wang et al., 2009).

In the first four years after 1996, the growth of online schools (at China) was fast, and the concepts of e-learning and online school were formulated. In the following four-year adjusting stage, the number of online schools decreased remarkably as some investors and managers of over-emphasized on financial gains but while ignoring Standards of web-based education. At the steady stage(2005-2009), managers and teachers have paid more attention to the educational role of online schools. They have developed better understanding and an improved model of cooperation. For example, one of the styles of cooperation is named “Two Instructors Cooperation Model” which means while one instructor teaching virtually, there is the other instructor (teaching assistant) working in the physical classroom to direct the students to dig more from the online instructions.

There are three models of running K12 online schools in China. Some online schools are run by a consortium of the government, or by an enterprise or a school alone. Most online schools in China are supported by private enterprises (Q. Wang et al., 2009). Online education, also known as virtual or cyber schooling, is a formal of distance education that uses the Internet and Computer technologies to connect teachers and students and deliver curriculum. Students may also communicate online with their classmates, students in other schools around the world and experts to whom they might otherwise not have access. Online learning may take the form of a single course for a student who accesses that course while sitting in a physical school, or it may replace the physical school for most or all of a student’s courses (Watson & Gemin, 2010).

K12 Online Education Market in China

In 1996, China’s first primary and secondary education website (online school), 101 online school, was established. Four year later, the Beijing No.4 Online School was built. The education at this time was mainly based on academic education, and the form was mostly based on a single form such as text mail. The interactive form was not strong, mainly based on traditional education methods (Chen et al., 2009). Students and online teachers could only use text to do interactive activities. There were no other choices for the online learners in China who wanted to get higher quality K-12 online instructions. The reasonable online instructions should be more interactive which requires the online instructors to communicate with the learners closely and share the feedback of the learners’ information with their parents frequently. Online instructors need to foster the online learners’ enthusiasm and interest in learning. It is also the online instructors’ duty to embed more technique tools and enhance their skills to improve their online teachings (Hu & Meyen, 2013).

Both Beijing 101 school and Beijing No.4 School are two of best high-schools in Beijing, China. (The author worked at Beijing No.4 School for ten years and has learnt that there were very few requirements for online instructors who taught at Beijing No.4 Online School.) In the past decade, online education market in China changed so fast. More and more companies like “Gen Shei Xue”, “Hao Wei Lai”, “New Oriental School”, etc. have stepped into the K-12 training market to set very high standard and expectations of online education. The parents would prefer to pay more money to buy better online education from these kinds of companies for they keep improving their online teaching strategies (Huang & Hong, 2017).

Quality of Online Teaching

To help China’s K-12 online instructors improve is the first step to get high-quality K-12 online education. Without clear and specific goals, the online instructors will lose themselves and cannot improve their online instructions highly (Baran et al., 2011). There are a lot of mature online teaching standards and suggestions written in English world which might be meaningful for Chinese colleagues.

In 2010, an international quality standard for e-learning programs – “Open ECBCheck” – was officially released. ECBCheck is an accreditation and quality improvement scheme for e-learning programs which supports organizations in measuring the success of their programs and allows for continuous improvement though peer collaboration. It was developed through an innovative and participative process involving more than 40 international, regional and national capacity-development organizations (*Open ECBCheck – Quality Improvement Scheme for E-Learning Programmes | GIZ Global Campus 21, n.d.*).

In the standards (*E-Learning Methodologies – A Guide for Designing and Developing e-Learning Courses*, n.d., p. 14), the quality of an e-learning course is enhanced by: (1) Learner-centered content: E-learning curricula should be relevant and specific to learners’ needs, roles and responsibilities in professional life. Skills, knowledge and information should be provided to this end. (2) Granularity: E-learning content should be segmented to facilitate assimilation of new knowledge and to allow flexible scheduling of time for learning. (3) Engaging content: Instructional methods and techniques should be used creatively to develop an engaging and motivating learning experience. (4) Interactivity: Frequent learner interaction is needed to sustain attention and promote learning. (5) Personalization: Self-paced courses should be customizable to reflect learners’ interests and needs; in instructor-led courses, tutors and facilitators should be able to follow the learners’ progress and performance individually.

The third edition of American National Standards for Quality Online Teaching purposely provided the K-12 online and blended learning community with an updated set of openly licensed standards to help evaluate and improve online courses, online teaching and online programs with the contribution from another two documents known collectively as the : *American National Standards for Quality Online Programs* and *American National Standards for Quality Online Courses* (Powell et al., n.d. P5). The set of standards served to inform the team, allowing them to make community and research supported updates. Subsequently, author took time to evaluate each standard and found they met the following criteria: measurable, valid, complete, relevant, and specific. Chinese online instructors can check the standards with clear definitions and explanations with examples to transfer to their online teaching actions.

Besides the formal standards which can help online instructors clarify their to-do lists about how to provide appropriate online instructions, the suggestions from some institutes are useful to bring more insights to improve online education for K-12 online instructors, especially during the COVID-19 pandemic spread period. The Danielson Group Remote Teaching Guide (*Danielson Group Remote Teaching Guide.Pdf*, n.d.) is one of them. Each academic year, the subject teachers in the author′ school will provide at least two class periods to be observed by the members in Academic Department to show their teaching abilities which we name them formal observations. The Academic Department members will use the Framework (Danielson, 2014) for teaching from the Danielson group for assessing subject teachers′ formal observations. The Remote Teaching guide is reflected in Framework for Teaching which shared eight suggestions to online instructors from knowing and valuing the online learners to build responsive learning environment, then to engaging students in learning. The online instructors will find more valuable strategies and tips to miss overwhelming when they conduct online instruction if they delve into deeper with reading and practice the suggestions in the guide.

Trainings for Enhancing Instructors′ Online Teaching Competencies

As online learning grows in K-12, the need to prepare quality online instructors increases (Borup & Evmenova, 2019). There are a lot of barriers the quality online K-12 instructors need to learn to cross. The barriers between the good online K-12 classroom and low efficiency which were listed two decades ago still affect online instructors. The barriers are: Academic, Fiscal, Geographic, Governance, Labor-Management, Legal, Student support, Technical, Cultural (Berge & Mrozowski, 1999). If the teachers did not learn how to face to the barriers when they were in the college, they may get some in-service professional development (PD) programs to help them add this part of competencies. If the PD programs are held fluently, the learners may improve their online teaching skills and knowledge (first-order barriers to change) and nurturing positive attitudes and dispositions (second-order barriers to change) (Borup & Evmenova, 2019). The online PD itself is a good model to show the learners about what is a good online course you need to learn to conduct. There are a lot of difference between face-to-face teaching and online teaching. Barbour (2012) found that the different competencies were required to deliver high quality online teaching. Even the experienced face-to-face teachers might fail to present quality online instructions. The effective PD program will deliver content and assignment proved effective at increasing faculty members′ knowledge and skills, but it was the course delivery and the opportunity to learn as an online student that appeared to most impact faculty members′ attitudes and perceptions of what was possible in online learning environments. In other words, the method was just as important as the message. When designing professional development courses, universities (or other departments) not only need to consider what will be learned but how it will be learned. If courses do not model effective online instruction, they run the risk of increasing faculty members' skills without improving their practice (Borup & Evmenova, 2019). There are some successful PDs like Borup and Evemenonva's PD (2019) which covers the following topics: (1) course design and development, (2) assessment and feedback, (3) student collaboration, (4) discussions, and (5) presence and support.

Methodology

- Mixed-Method

The mixed-methods will be used. While conducting a bibliometrics focus on online teaching, hybrid teaching, online education standards, etc. The information will be concluded and compared to be selected as suggestions for assisting Chinese K-12 online teachers in their trainings and in-service instructions. Then, some questionnaires will be distributed to online educators, learners and parents to get their feedback about online instructions. The data will be used by qualitative and quantitative methods to test which suggestions or standards will be useful to the online instruction stake holders.

- Research Model

- Brief rationale for the selected approach

With the qualitative and quantitative research, the researcher can collect more divisions of feedbacks about online instructions from online teachers, learners and parents. The aims of the research are figuring out the most reasonable and workable assistance to Chinese K-12 online instructors. We hope to stand on the giant's shoulder and face to the reality of China's K-12 online education to compose two documents: *Chinese Online Instruction Standards*, and *Chinese Online Instruction Suggestions*, besides that, we will develop PD programs to assist Chinese K-12 online instructors. We wish they are reasonable and meet China's online instructors' expectations. Due to the different cultural and historic backgrounds in China and other countries or districts, the experiences from others must be improved to be embedded in Chinese education context. That will be the researcher's following research aims.

Variables

Independent variables: online teaching standards with using or not; online teaching suggestions with delivering or not; Weather took part in the PD for assisting online instructors developed by researcher.

Dependent variables: Online teachers' instructional performance scores with using online teaching standards to evaluate; Teachers' online instruction materials establishing time using before and after training; Teachers' degree of satisfaction of online instruction before and after training;

Online learners’ degree of satisfaction of online education; Parents’ degree of satisfaction with online education.

Sample

170 online learners from primary school, secondary school and high school.

50 online teachers from Future Leadership Academy (a private K-12 school).

100 college students who want to be teachers after graduation at Beijing Normal University.

100 in-service teachers in China K-12 schools.

100 parents whose kids are having online courses.

Data Collection

Qualitative data: Survey, interviews

Quantitative data: Survey, formal observations, data record on learning management system, interviews

Ethical Considerations

Students and parents’ privacy protection. The data from the students and parents will be ONLY used for research and will not be open to others without erasing the names of the students and parents.

Data Analysis

Correlation Analysis using SPSS

Timeline

February, 2021. Preparation period:

In the coming school-wide professional development program, the research proposal will be informed.

February, 2021. Transition period:

the online instructors will be required to having online teaching trainings. The suggestions and strategies will be shared with them as well.

March, 2021. Orientation period:

1. The online learners and parents will be delivered online education orientation with introducing more details of online courses and suggestions for online learnings.
2. The online instructors will be directed to have online trainings with sharing more online teaching standards.

April-June 2021. Formal online instructions conduction

1. Data collection.
2. Survey
3. Interviews to teachers, students and parents.

July 2021. Data analysis and reaching a conclusion

References

Baran, E., Correia, A.-P., & Thompson, A. (2011). Transforming online teaching practice: Critical analysis of the literature on the roles and competencies of online teachers. *Distance Education*, *32*(3), 421–439. https://doi.org/10.1080/01587919.2011.610293

Barbour, M. k. (2012). Training teachers for a virtual school system: A call to action. In *Developing Technology-Rich Teacher Education Programs: Key Issues*. IGI Global.

Berge, Z. L., & Mrozowski, S. E. (1999). *Barriers to Online Teaching in Elementary, Secondary, and Teacher Education*. https://doi.org/10.21432/T2P59B

Borup, J., & Evmenova, A. (2019). The Effectiveness of Professional Development in Overcoming Obstacles to Effective Online Instruction in a College of Education. *Online Learning*, *23*(2), Article 2. https://doi.org/10.24059/olj.v23i2.1468

Carliner, S., & Shank, P. (2008). *The e-learning handbook: Past promises, present challenges* (MU Ellis Library). Pfeiffer.

Chen, L., Wang, N., & Qiao, A. (2009). K12 online school practice in China. *Campus-Wide Information Systems*, *26*(2), 137–144. https://doi.org/10.1108/10650740910946864

Clark, R. C., & Mayer, R. E. (2011). *E-learning and the science of instruction: Proven guidelines for consumers and designers of multimedia learning* (MU Online; 3rd ed.). Pfeiffer.

Danielson, C. (2014). *The framework for teaching: Evaluation instrument*.

Danielson Group Remote Teaching Guide.pdf. (n.d.).

E-learning methodologies—A guide for designing and developing e-learning courses. (n.d.). 141.

Growth of China's Online Education Industry Spurs New Regulations. (2020, March 12). China Briefing News. https://www.china-briefing.com/news/growth-chinas-online-education-industry-spurs-new-regulations/

Hu, X. C., & Meyen, E. L. (2013). A Comparison of Student and Instructor Preferences for Design and Pedagogy Features in Postsecondary Online Courses. In *Curriculum, Learning, and Teaching Advancements in Online Education*. IGI Global.

Huang, X., & Hong, W. (2017). Live Streaming Teaching Applied in Real English Classroom. *DEStech Transactions on Social Science, Education and Human Science*, *0*(aems), Article aems. https://doi.org/10.12783/dtssehs/aems2017/8278

King, K. P. (2002). Educational technology professional development as transformative learning opportunities. *Computers & Education*, *39*(3), 283–297. https://doi.org/10.1016/S0360-1315(02)00073-8

Open ECBCheck – Quality improvement scheme for E-Learning programmes | GIZ Global Campus 21. (n.d.). Retrieved October 28, 2020, from https://gc21.giz.de/ibt/var/app/wp342P/1522/index.php/open-ecbcheck/

Powell, A., Oliver, W., Archambault, L., Borup, J., Boyarko, M., Cavanaugh, C., Copeland, R., Day, J., DeRosa-Davis, T., Deschaine, M., Graham, C., Hanson, E., Hernandez, B., Imbrunone, M., Kennedy, K., Consulting, K. K., Kolar-Burden, J., Little, T., Loomis, K., … Online, P. (n.d.). *National Standards for Quality Online Teaching*. 33.

Wang, Q., Zhu, Z., Chen, L., & Yan, H. (2009). E-learning in China. *Campus-Wide Information Systems*, *26*(2), 77–81. https://doi.org/10.1108/10650740910946783

Wang, Y., Liu, X., & Zhang, Z. (2018a). An overview of e-learning in China: History, challenges and opportunities. *Research in Comparative and International Education*, *13*(1), 195–210. https://doi.org/10.1177/1745499918763421

Wang, Y., Liu, X., & Zhang, Z. (2018b). An overview of e-learning in China: History, challenges and opportunities. *Research in Comparative and International Education*, *13*(1), 195–210. https://doi.org/10.1177/1745499918763421

Watson, J., & Gemin, B. (n.d.). *A Parent's Guide to Choosing the Right Online Program*. 23.

— — . (n.d.). Retrieved October 28, 2020, from http://www.ict.edu.cn/news/n2/n20150525_24914.shtml